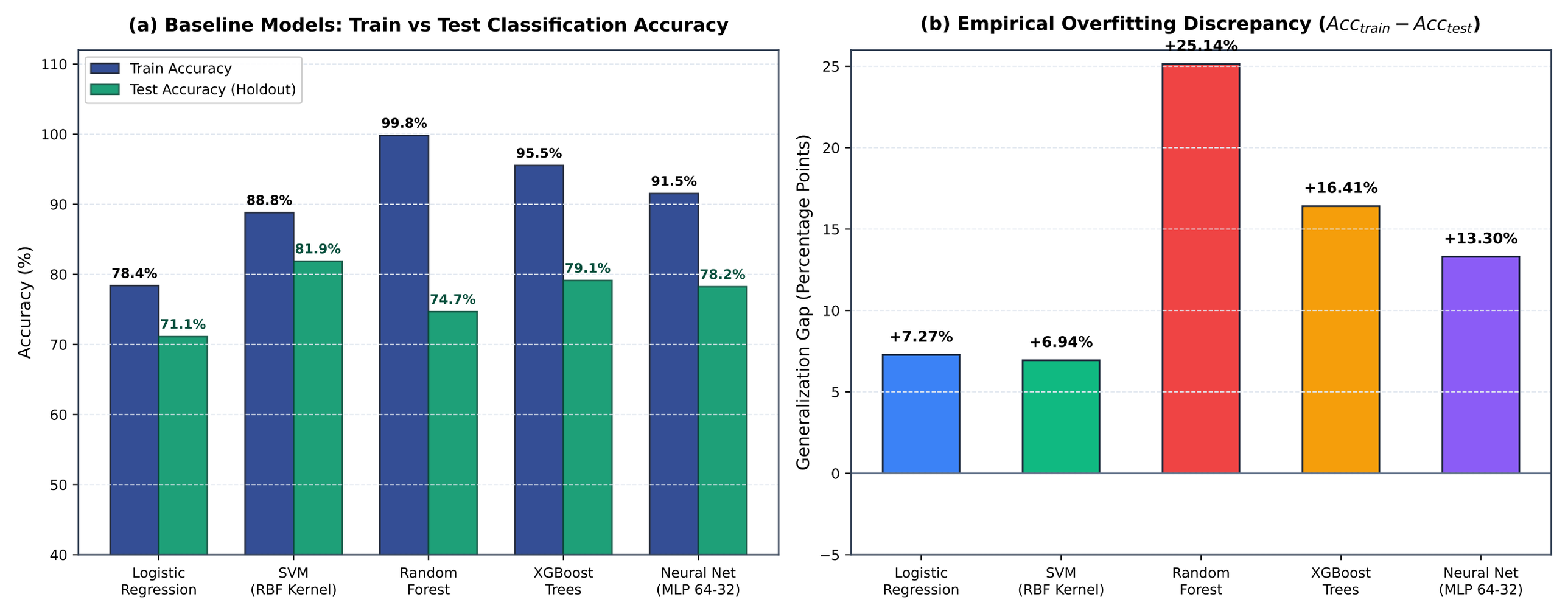
(a) Baseline Models: Train vs Test Classification Accuracy
Train Accuracy
Test Accuracy (Holdout)
Accuracy (%)
110
100
90
80
70
60
50
40
78.4%
71.1%
88.8%
81.9%
99.8%
74.7%
95.5%
79.1%
91.5%
78.2%
Logistic Regression
SVM (RBF Kernel)
Random Forest
XGBoost Trees
Neural Net (MLP 64-32)
(b) Empirical Overfitting Discrepancy ($Acc_{train} - Acc_{test}$)
Generalization Gap (Percentage Points)
25
20
15
10
5
0
−5
+7.27%
+6.94%
+25.14%
+16.41%
+13.30%
Logistic Regression
SVM (RBF Kernel)
Random Forest
XGBoost Trees
Neural Net (MLP 64-32)